



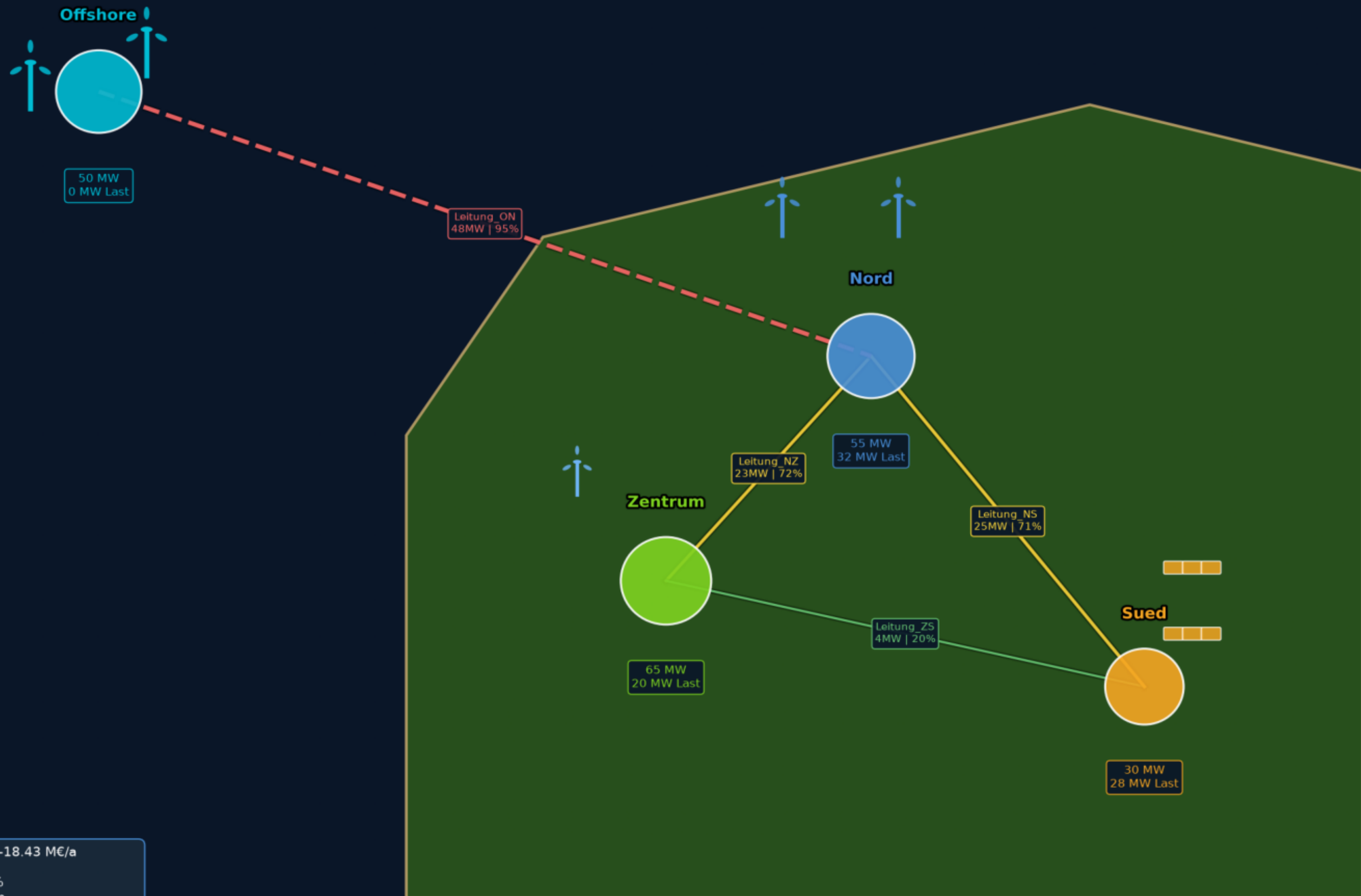
Island Energy Model v3.0

4-Zonen-Inselmodell mit Sektorkopplung,
H₂-System & Monte-Carlo-Robustheitsprüfung

Gesamtkosten: -18.426 M€/a | CO₂: 2 tCO₂/a | Zeitschritte: 8760

Erstellt: 19.06.2026 23:52

Ausgabe: d:\Project_PyPSA\Lummerland_Output

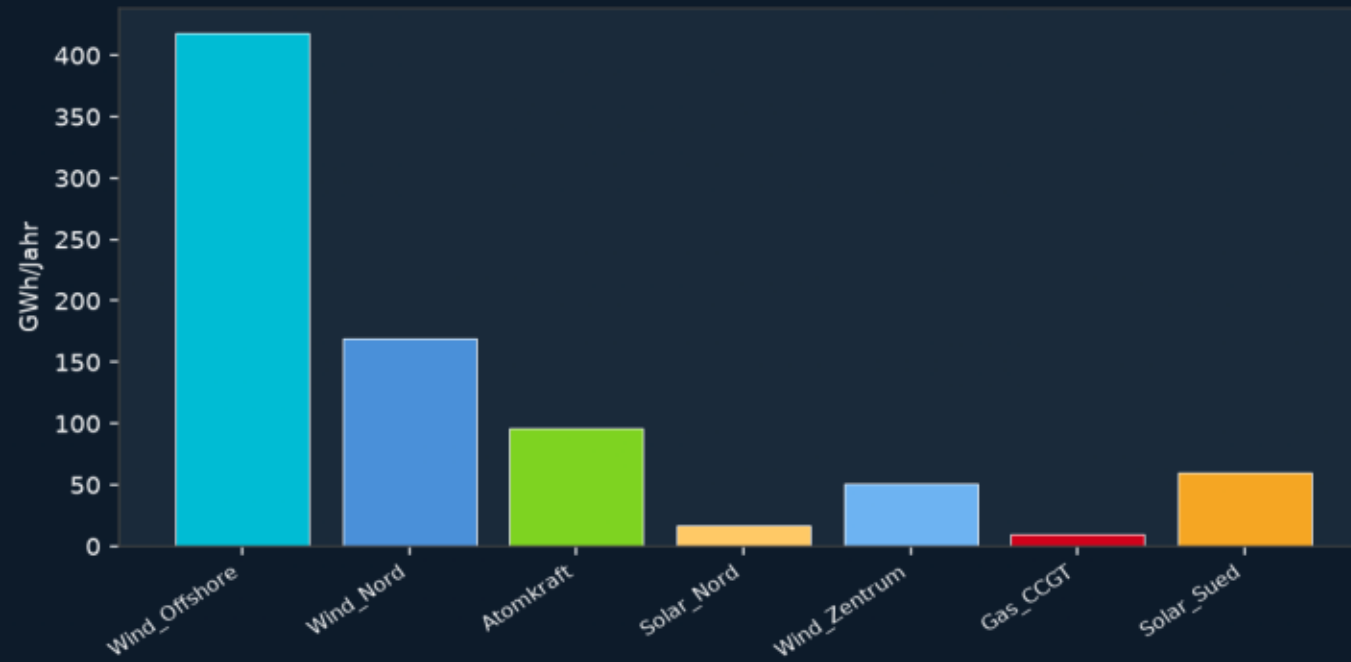


Gesamtkosten: -18.43 M€/a
 CO₂: 2 t/a
 RE-Anteil: 87.1%
 H₂-Tank: 50 MWh
 Elektrolyseur: 2 MW
 Brennstoffzelle: 2 MW
 WP Zentrum: 13 MW | WP Süd: 9 MW

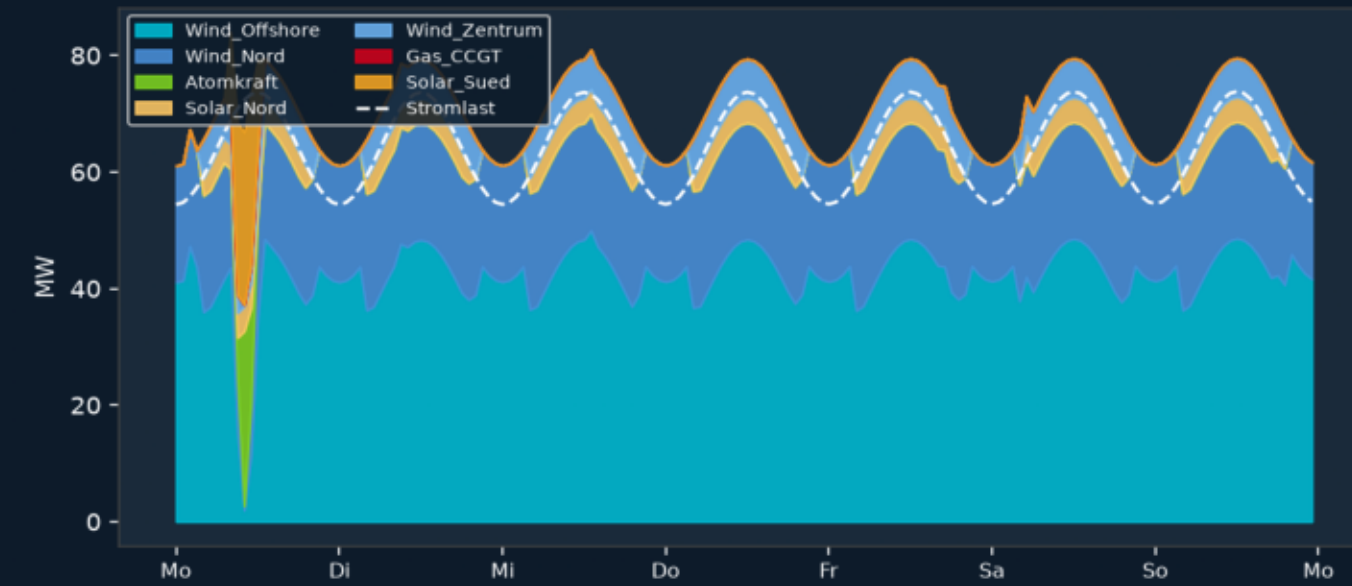
<50%
 50-80%
 >80%
 Offshore

LUMMERLAND v3.0 - Erzeugungsmix, H₂-System & Sektorkopplung

Jährlicher Erzeugungsmix



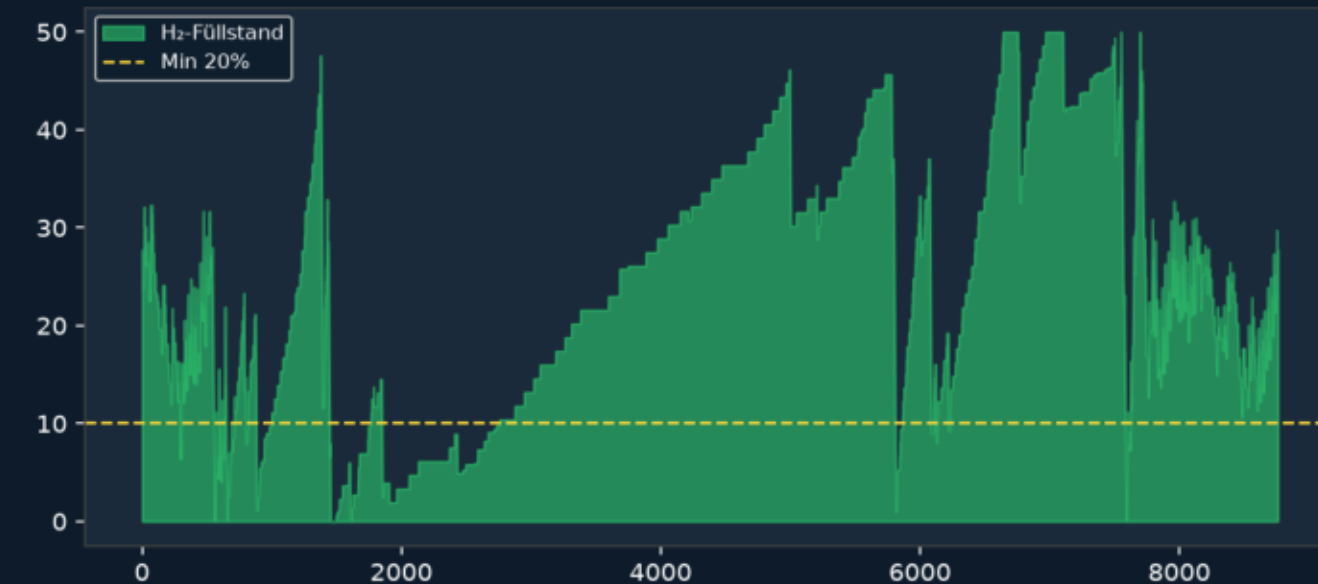
Sommer (KW 26)



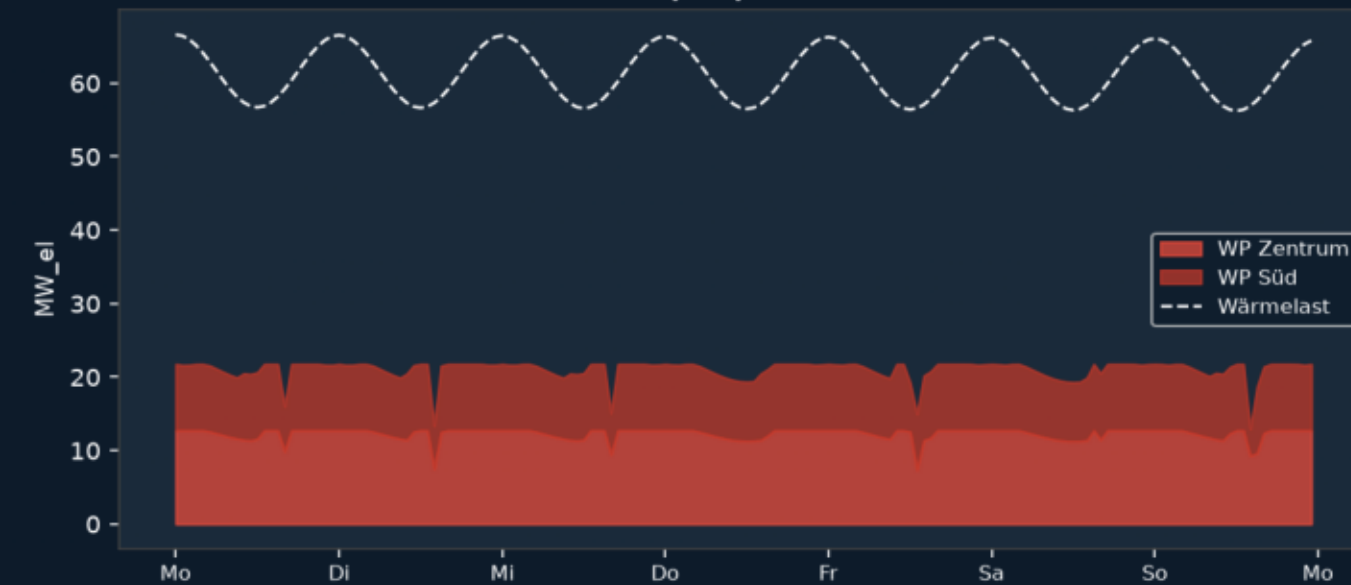
Winter (KW 1)



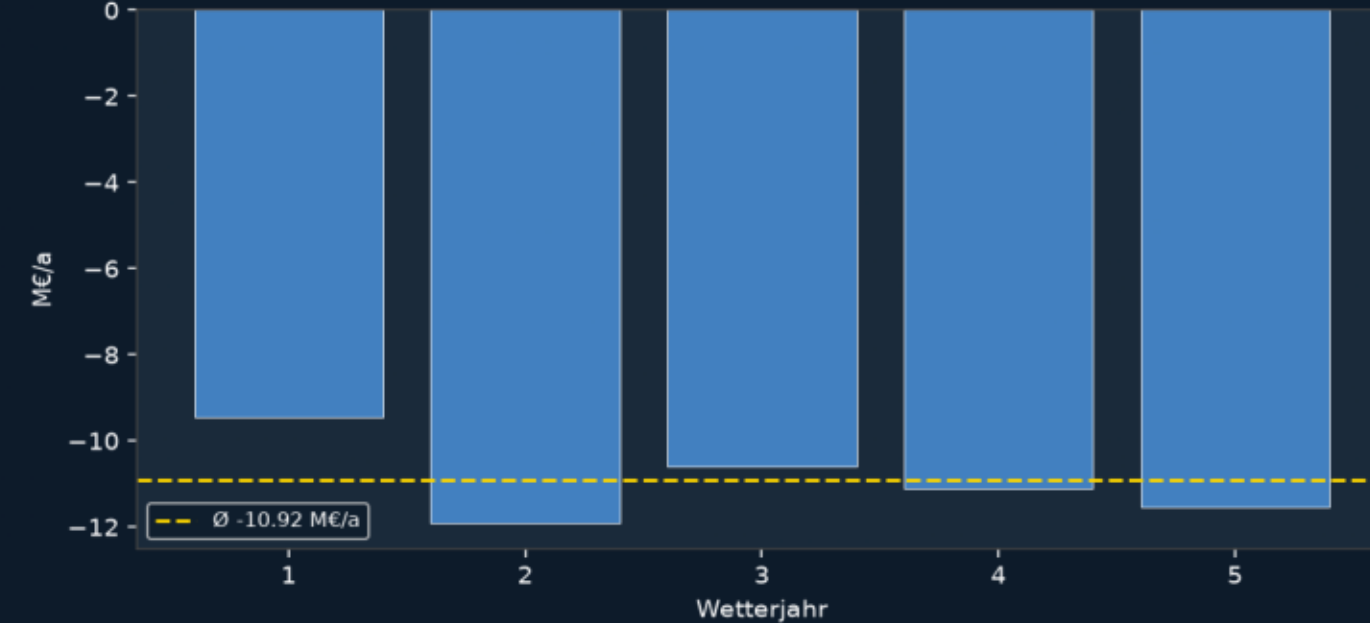
H₂-Tank SoC



Wärmepumpen - Winter

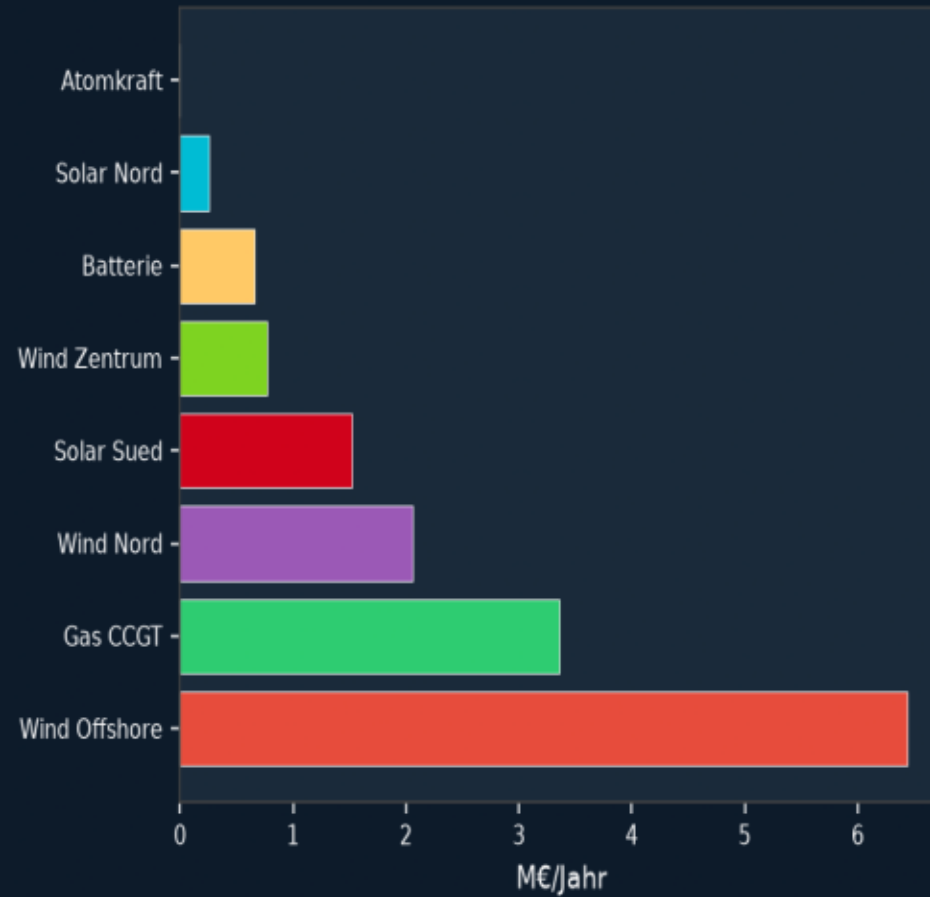


Monte-Carlo Kosten

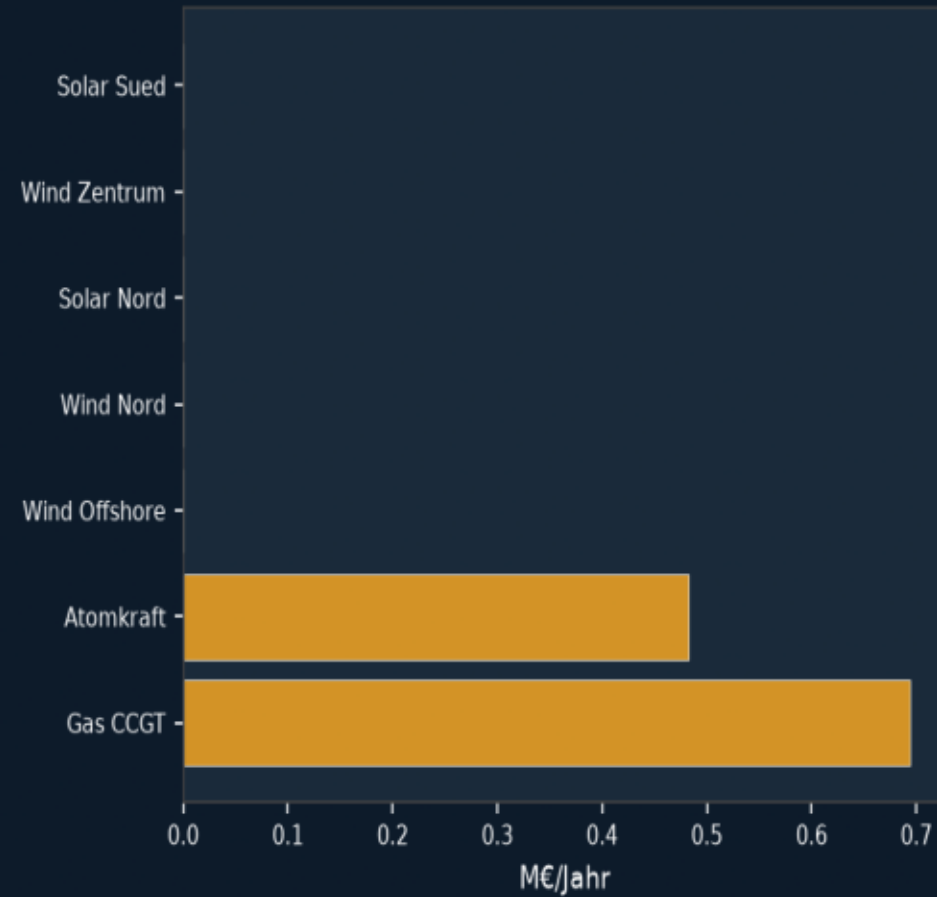


LUMMERLAND v3.0 - Kosten, CO₂ & Monte-Carlo Robustheit

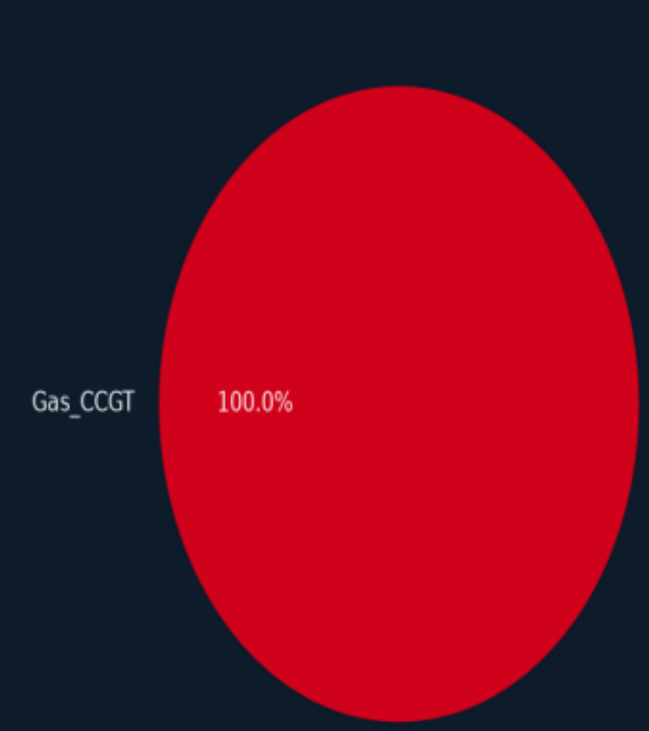
Kapitalkosten



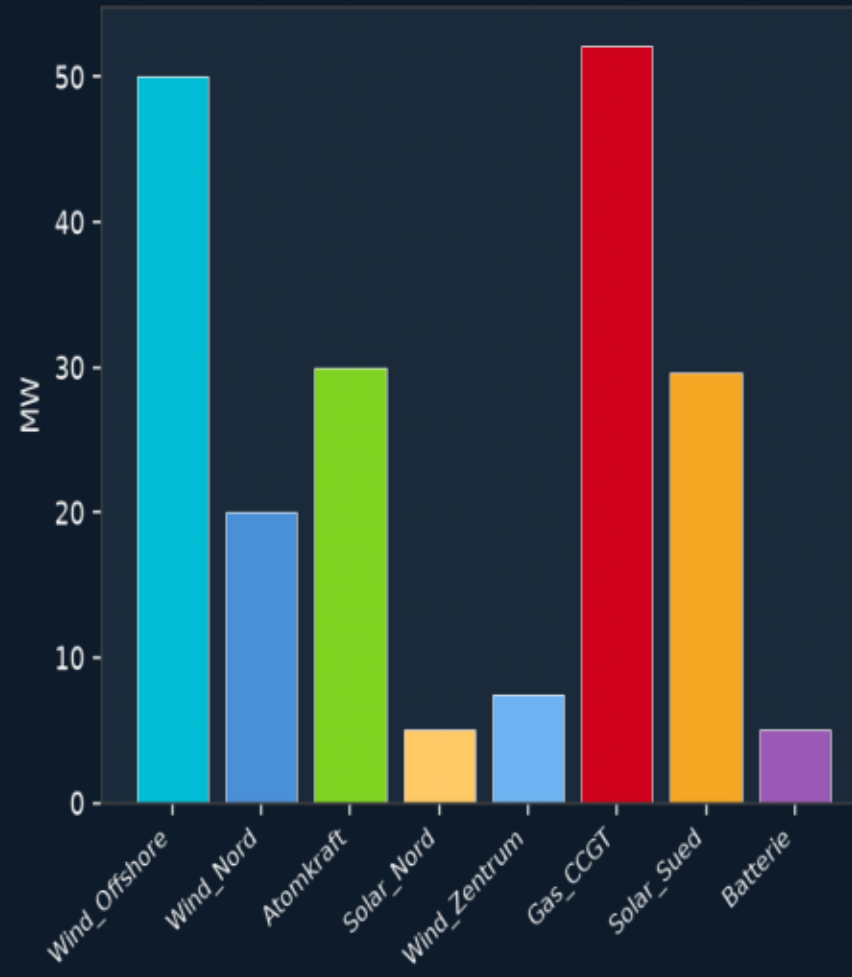
Betriebskosten



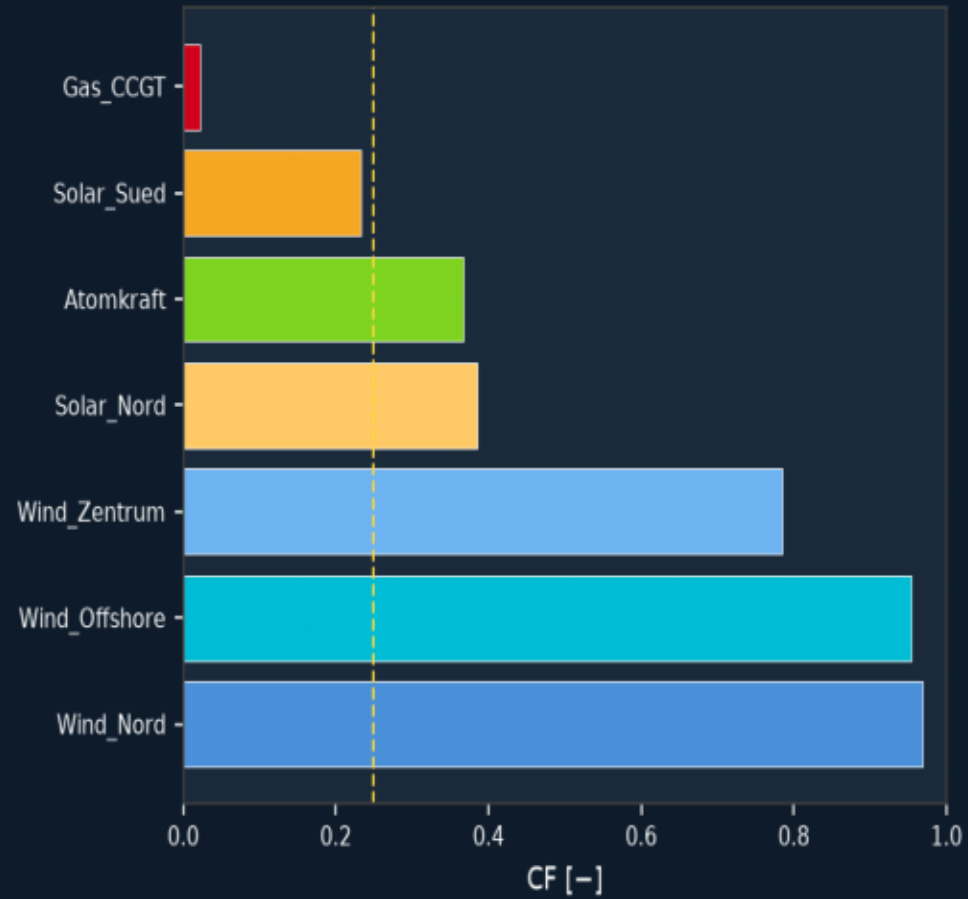
CO₂-Emissionen (Budget: 20,000 t/a)



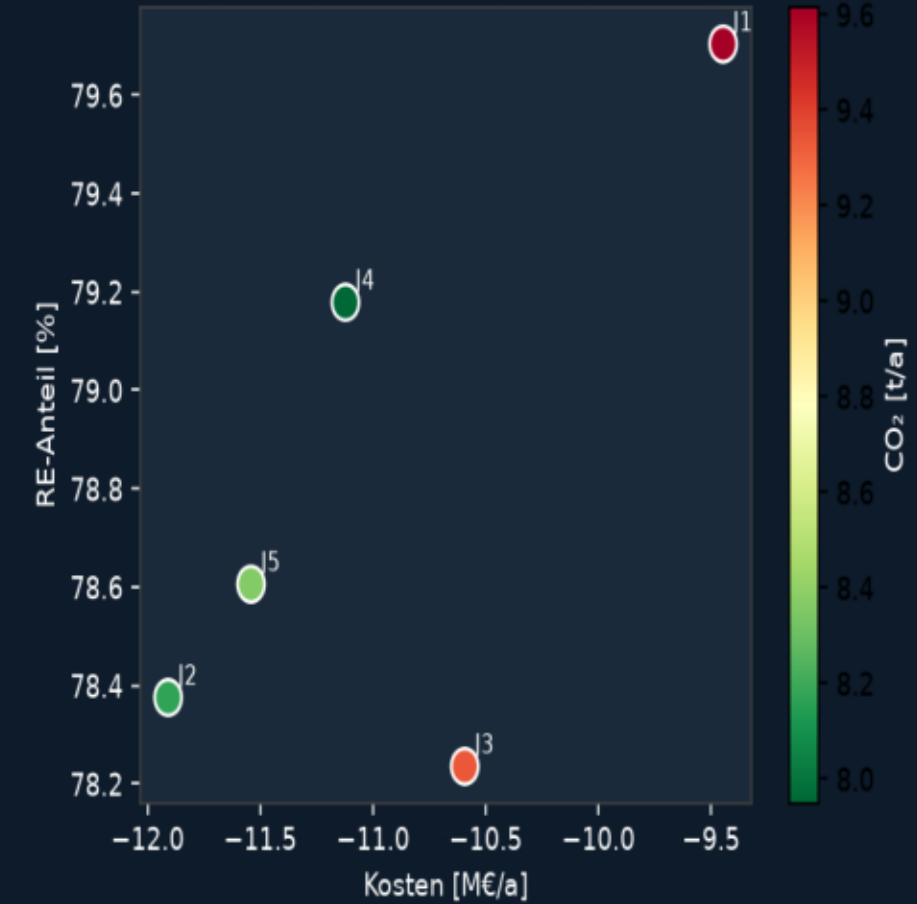
Optimierte Kapazitäten



Kapazitätsfaktoren

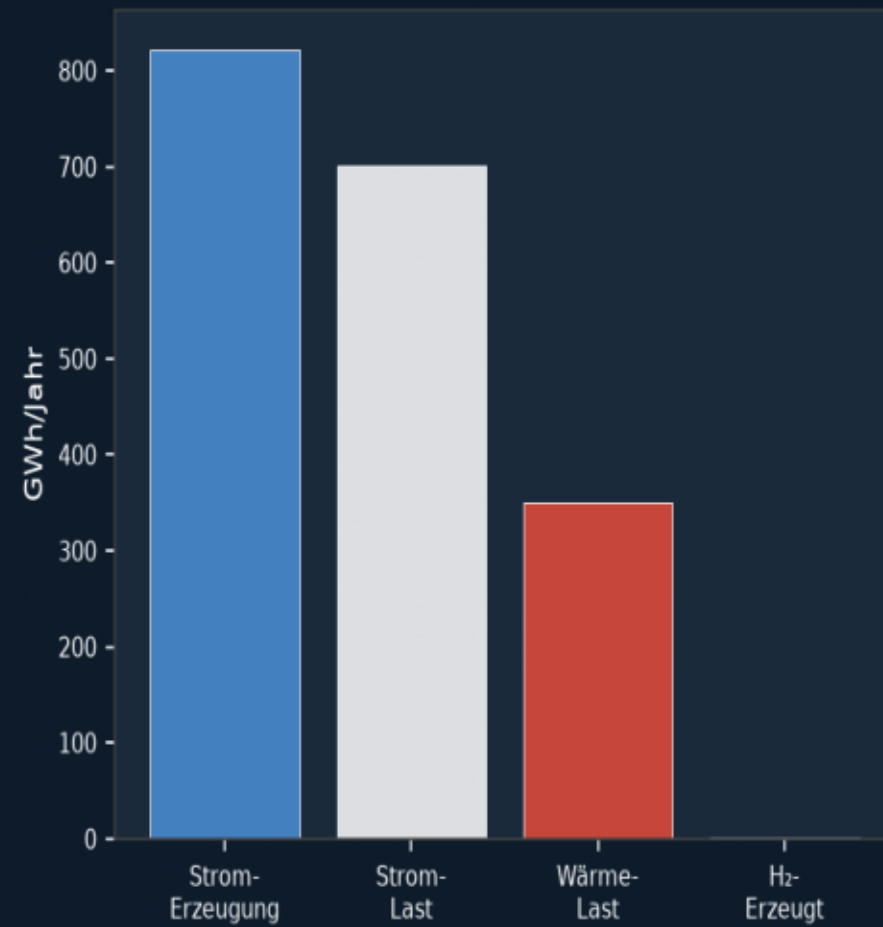


MC: Kosten vs. RE-Anteil

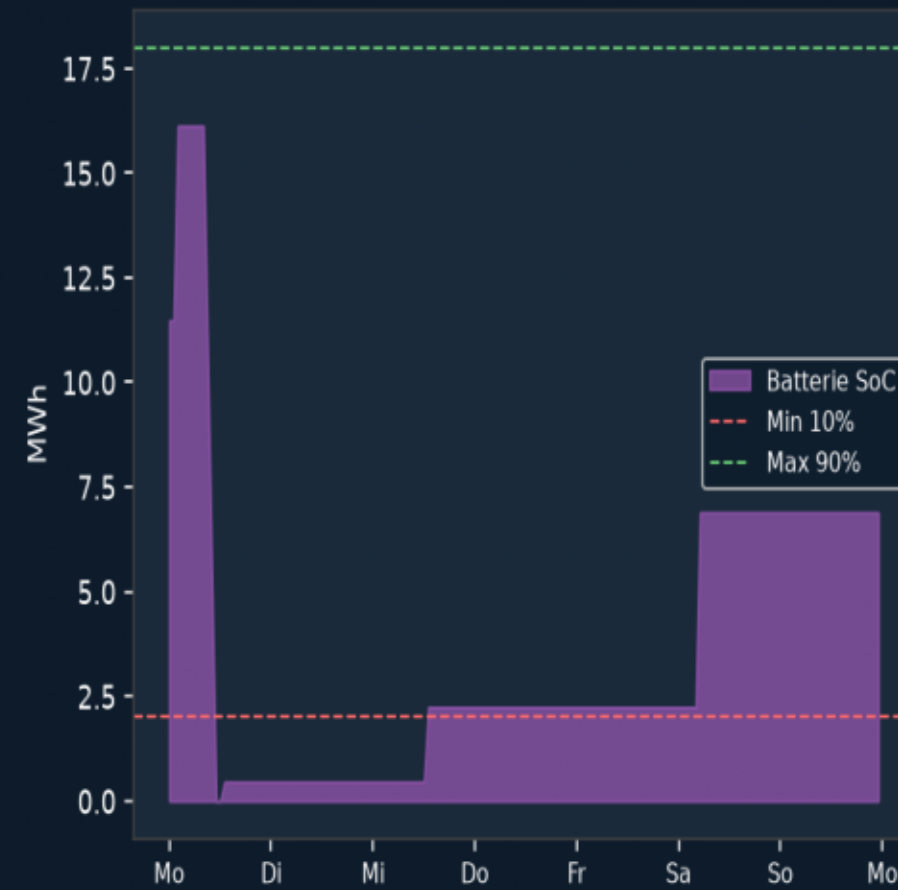


LUMMERLAND v3.0 - Sektorkopplung, Speicher & Versorgungssicherheit

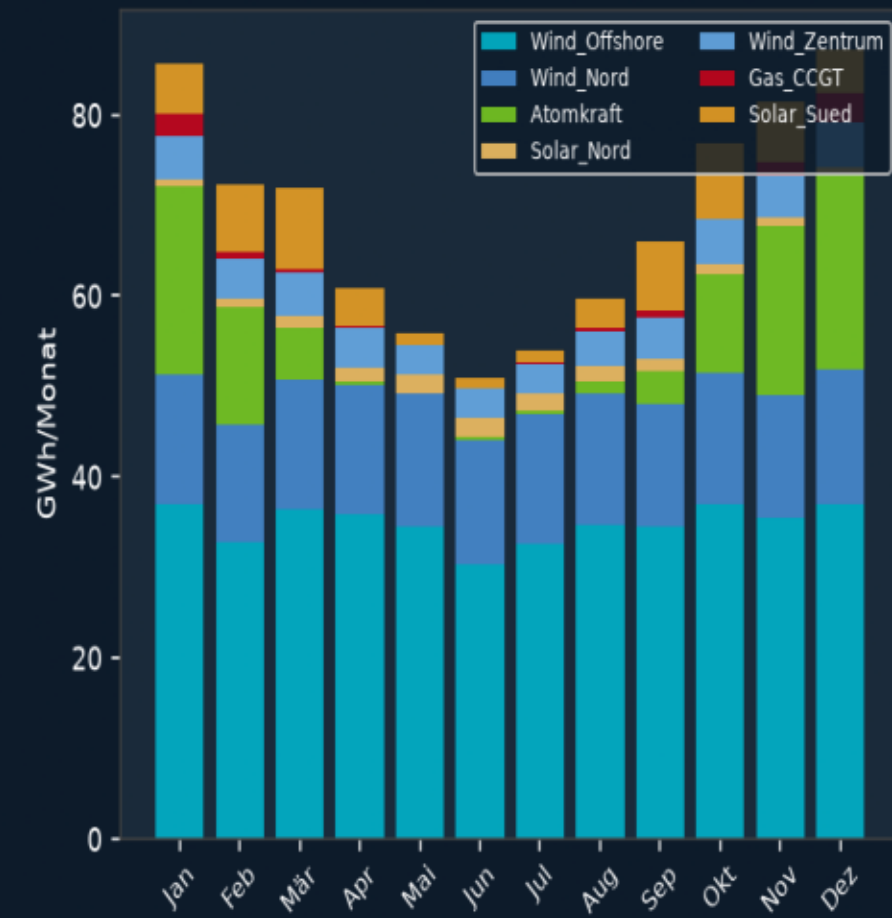
Jahresenergie-Übersicht



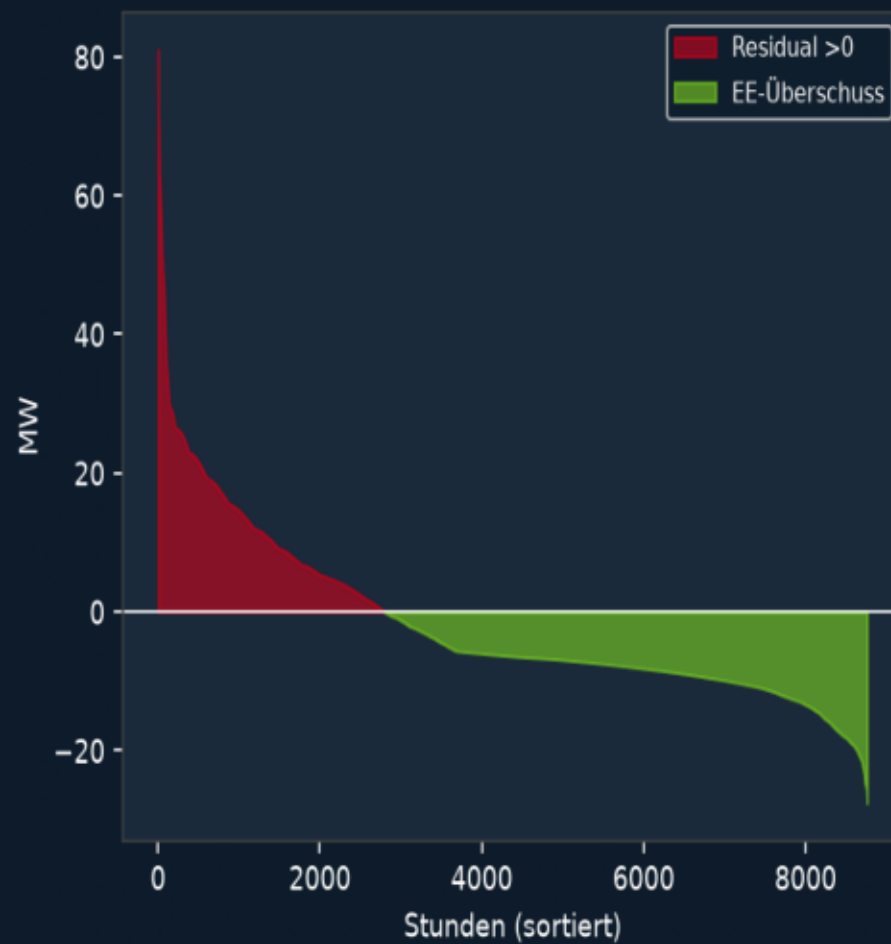
Batterie SoC - Sommer



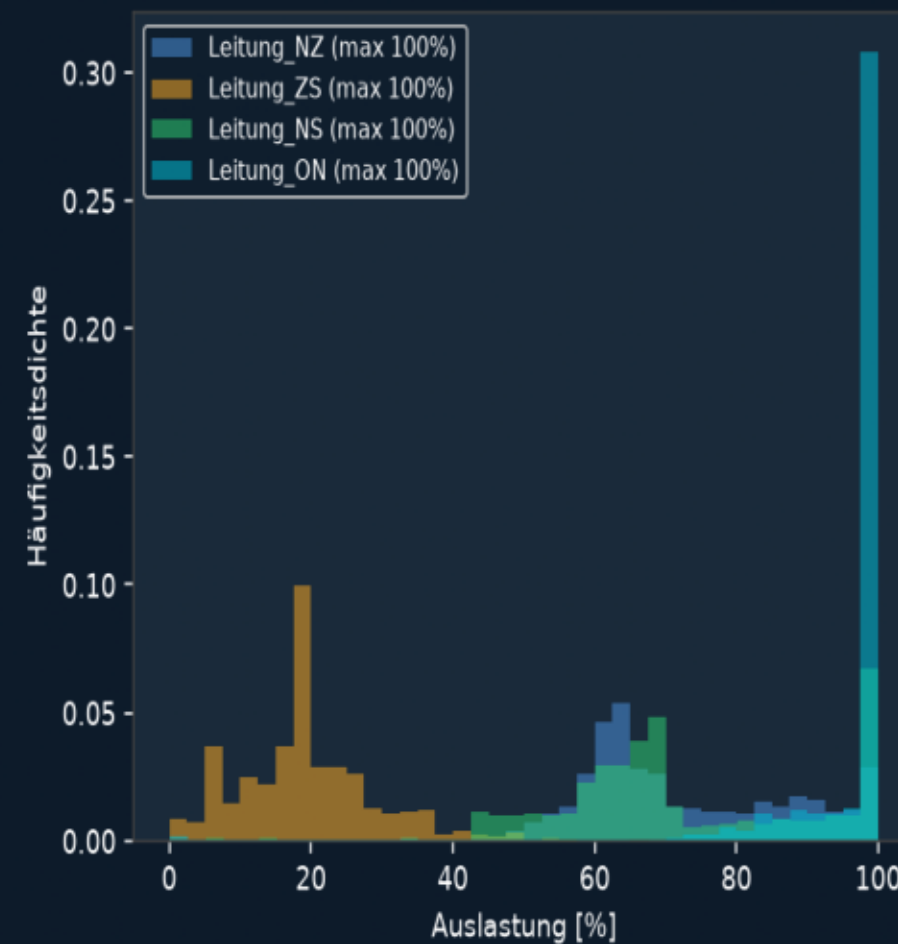
Monatliche Erzeugung



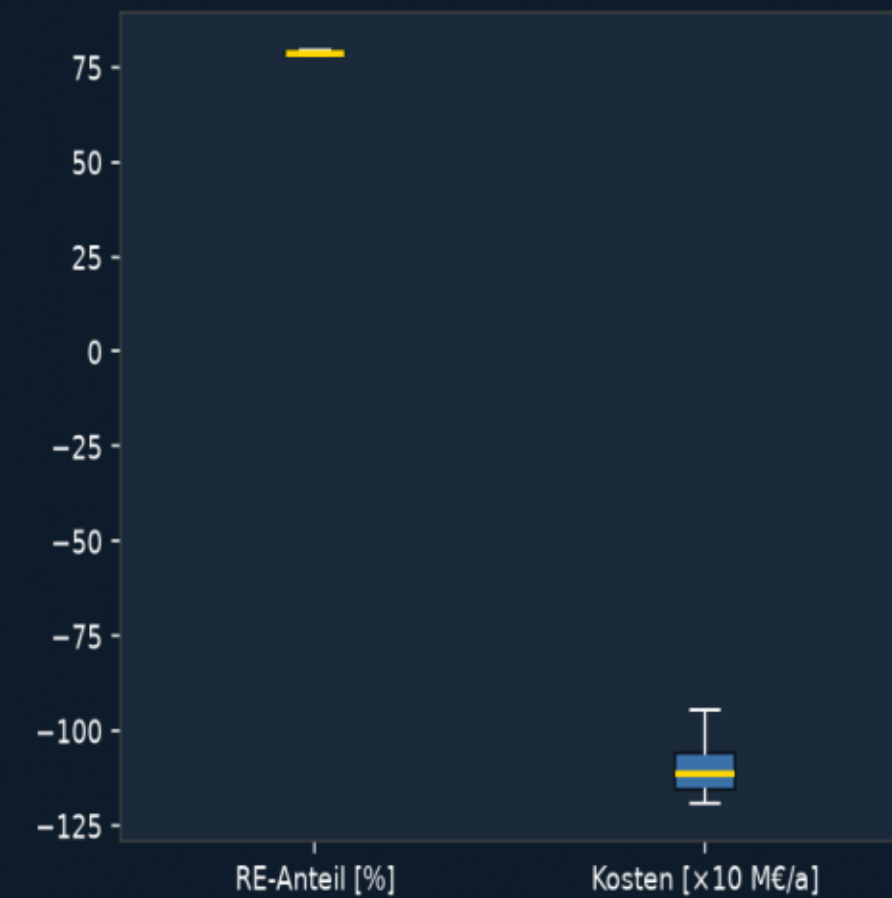
Dauerlinie Residuallast



Leitungsauslastung



Monte-Carlo Streuung



Monte-Carlo Robustheitsprüfung

Jahr	Kosten [M€/a]	CO ₂ [t/a]	RE-Anteil [%]	Status
1	-9.45	10	79.7	OK
2	-11.91	8	78.4	OK
3	-10.59	9	78.2	OK
4	-11.12	8	79.2	OK
5	-11.54	8	78.6	OK

Ø Kosten: -10.92 M€/a | Ø CO₂: 9 t/a | Ø RE: 78.8%